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The healthiest competition
occurs when average people
win by putting above average
effort.

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Good

Evening

Today's content

01. Nearest smaller on left/right
02. Nearest greater on left/right
03. Largest Area histogram
04. Sum of (max-min) in all subarrays

01. Given an integer A[]: Find the nearest smaller element index on left.

A[] = {⁰4 ¹5 ²2 ³10 ⁴3 ⁵12}

Ans[] = {-1 0 -1 2 2 4}

A[] = {⁰8 ¹2 ²4 ³9 ⁴7 ⁵5 ⁶3 ⁷10}

Ans[] = {-1 -1 1 2 2 2 1 6}

A[] = {⁰4 ¹6 ²10 ³11 ⁴7 ⁵8 ⁶3 ⁷5}

Ans[] = {-1 0 1 2 1 4 -1 6}

A[] = {⁰4 ¹5 ²2 ³10 ⁴8 ⁵2}

Ans[] = {-1 0 -1 2 2 -1}

Brute force → For an element at ith idx, iterate from i-1 to 0 & look for the nearest smaller element.

```
int[] ans = new int[n];
```

```
for (i=0; i<n; i++) {
```

```
    ans[i] = -1;
```

```
    for (j=i-1; j>=0; j--) {
```

```
        if (A[j] < A[i]) {
```

```
            ans[i] = j;
```

```
            break;
```

```
    }
```

```
return ans;
```

TC: $O(n^2)$

SC: $O(1)$

Observations

$$A[] = \{ 8, 2, 4, 9, 7, 5, 3, 10, 10, 11 \}$$

Can idx 0's ele (8) ever be the answer for x ?

01. $x \leq 5 \Rightarrow \text{ans} \neq 0^{\text{th}} \text{idx}$

02. $5 < x \leq 8 \Rightarrow \text{ans} = 4^{\text{th}} \text{idx} \quad (\text{ans} \neq 0^{\text{th}} \text{idx})$

03. $x > 8 \Rightarrow \text{ans} = 4^{\text{th}} \text{idx} \quad (\text{ans} \neq 0^{\text{th}} \text{idx})$

Conclusion $\rightarrow$ If a smaller no. is present on RHS of a larger no. then this larger no. can never contribute in the answer

$$A[] = \{ 8, 2, 4, 9, 7, 5, 3, 10, 10, 11 \}$$

$$\text{Ans}[] = \{ -1, -1, 1, 2, 2, 2, 1, 6, 6, 8 \}$$

9	11
8	10
7	10
6	3
5	5
4	7
3	9
2	4
1	2
0	8

Original
stack

corresponding
element

```
Stack <I> st = new Stack <>();
```

```
int [] ans = new int[n]
```

```
for (i=0; i<n; i++) {
```

```
    while (st.size() > 0 && A[i] ≤ A[st.peek()]) {
```

```
        st.pop();
```

```
    }
```

```
    if (st.size() == 0) ans[i] = -1;
```

$T_c: O(n)$

```
    else ans[i] = st.peek();
```

$S_c: O(n)$

```
    st.push(i);
```

```
}
```

∴ i, find nearest smaller or equal ele idx on left.

$A[i] < A[st.peek()]$

∴ i, find nearest greater ele idx on left

$A[i] \geq A[st.peek()]$

updating popping condition in above code to

get the answer for these variations.

A person uses **Google Maps** to find the nearest restaurants and picks one based on its proximity. Unfortunately, after visiting, they realized that the restaurant didn't meet their expectations.

Task

Let's break it down with a simple example. You have a list of restaurants and their **ratings**. For each restaurant, we're going to find the next restaurant to the **right** on the list that's not just close but also has a **higher rating than the current one**. If there's no better option on the list, we'll say there's **none** available.

Problem

Given a sequence of restaurants listed on Google Maps with their ratings, create a tool that helps users discover the rating of the next higher-rated restaurant to the right for each listed establishment.

↓

	0	1	2	3	4	5	6
A[] = {	3	2	6	5	8	7	9}
Ans[] = {	2	2	4	4	6	6	-1}

* Next greater element on RHS

```
Stack <I> st = new Stack <>();
```

```
int [] ans = new int [n]
```

```
for (i = n-1 ; i ≥ 0; i--)
```

```
    while (st.size() > 0 && A[i] ≥ A[st.peek()]) {
```

```
        st.pop();
```

```
        if (st.size() == 0) ans[i] = -1;
```

```
        else ans[i] = st.peek();
```

```
        st.push(i);
```

Tc: O(n)

Sc: O(n)

3

↓

	0	1	2	3	4	5	6
A[] = {	3	2	6	5	8	7	9}
Ans[] = {	2	2	4	4	6	6	-1}

0	3
1	2
2	6
3	5
4	8
5	7
6	9

original Stack ↔ corresponding ele

$A[i] < A[\text{st.peek()}]$
not remove

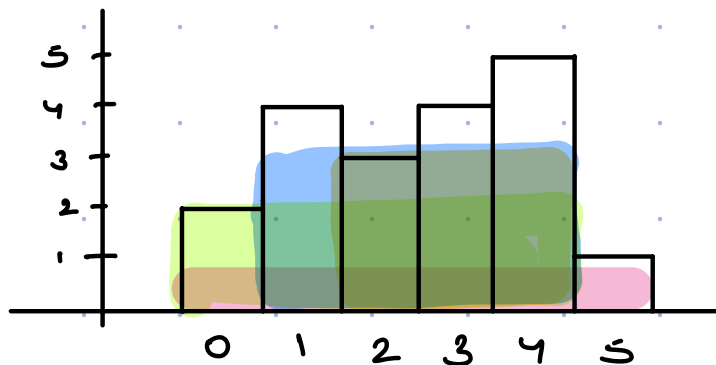
$\text{curr ele} \geq A[\text{st.peek()}]$
remove

Histogram Area

Given continuous block of Histogram find max Rectangular area which can be present within histogram

Note:- Every histogram is of width = 1

Ex : $arr[6] = \{ \underset{0}{2} \underset{1}{4} \underset{2}{3} \underset{3}{4} \underset{4}{5} \underset{5}{1} \}$



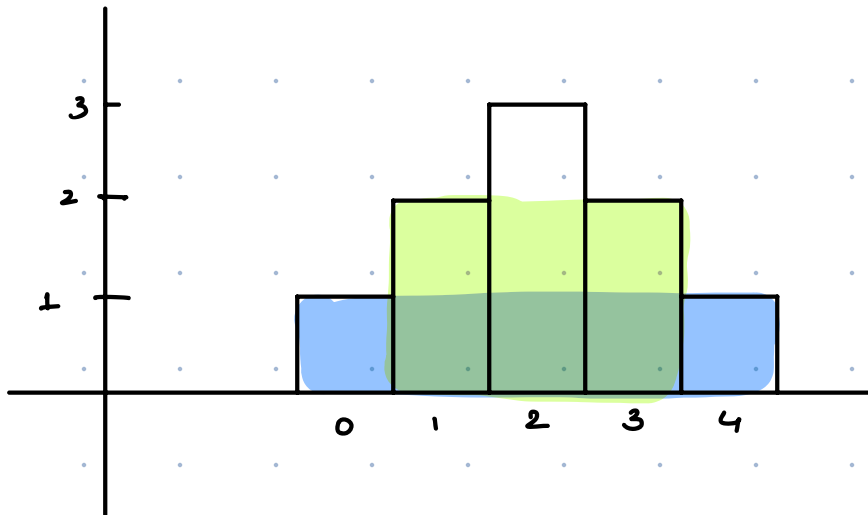
Area = 6

Area = 9

Area = 10

Area = 12 → find ans

$A[] = \{1 \ 2 \ 3 \ 2 \ 1\}$



Area = 5

Area = 2×3

Ans

Brute force $\rightarrow$ Consider all the possible combination of starting point & ending point of histogram

ans = 0

for ($i = 0$; $i < n$; $i++$) {

 minht = $A[i]$;

 for ($j = i$; $j < n$; $j++$) {

 minht = $\min(\text{minht}, A[j])$

 width = $j - i + 1$

 Area = $\text{minht} \times \text{width}$;

 ans = $\max(\text{ans}, \text{Area})$;

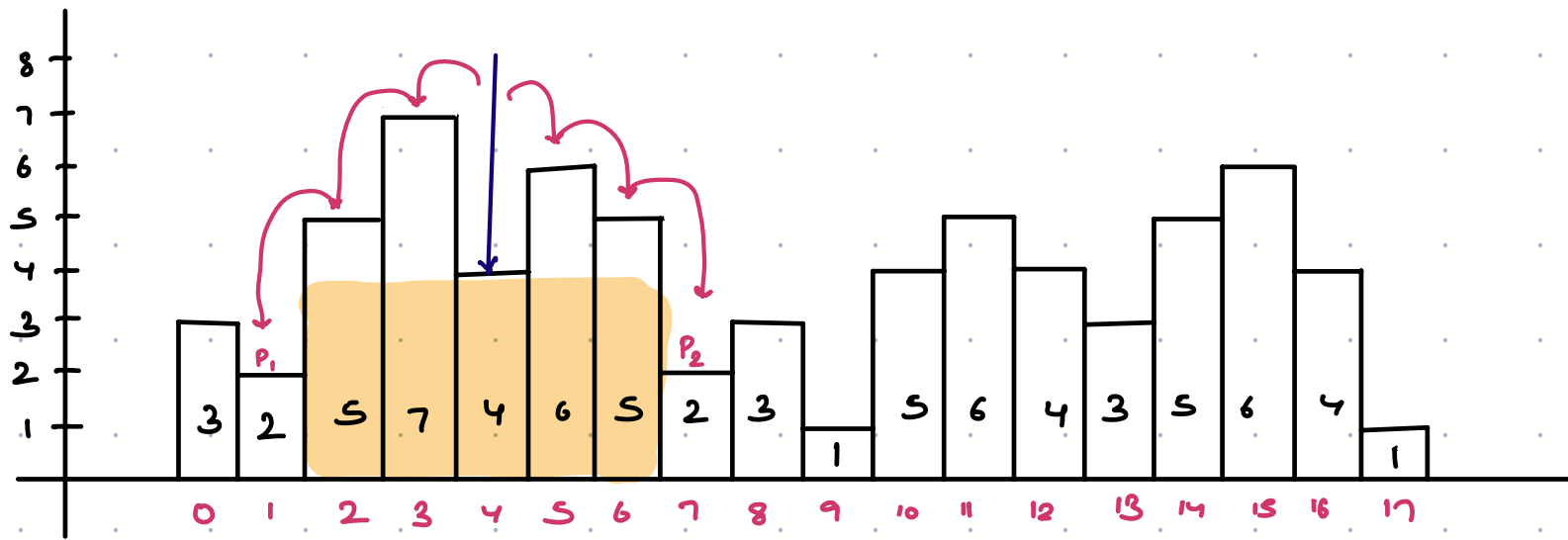
 }

Tc: $O(n^2)$

Sc: $O(1)$

arr[] =

3	2	5	7	4	6	5	2	3	1	5	6	4	3	5	6	4	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



Idea → Consider each histogram as height of my rectangle
 & figure out how much to expand on

LHS & RHS

Nearest smaller
 ele. idx on left

P_1

Nearest smaller ele
 idx on right.

P_2

$$\text{width} = P_2 - P_1 - 1$$

```
int rectangularArea ( int [] ht)
```

```
int [] left = nearestSmallerIdx on left(ht); // default val = -1
```

```
int [] right = nearestSmallerIdx on right(ht); // default val = n  
// & not -1
```

```
int area = 0, ans = 0
```

```
for (i=0; i<n; i++){
```

```
    int h = ht[i];
```

```
    P1 = left[i]
```

```
    P2 = right[i]
```

```
    area = h * (P2 - P1 - 1)
```

```
    ans = Math.max(ans, area)
```

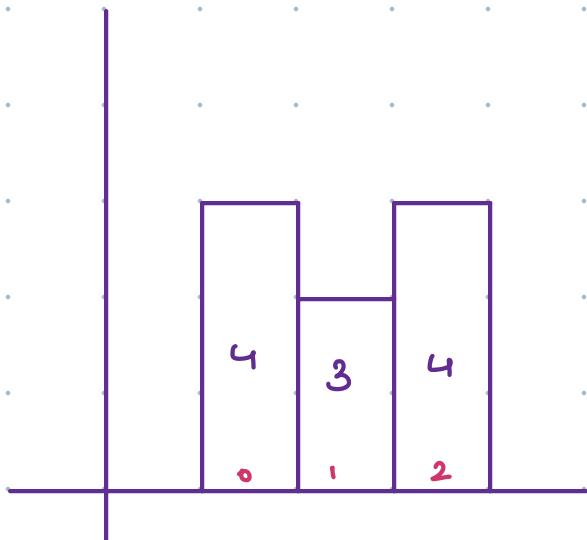
```
}
```

```
return ans;
```

3

TC: O(n)

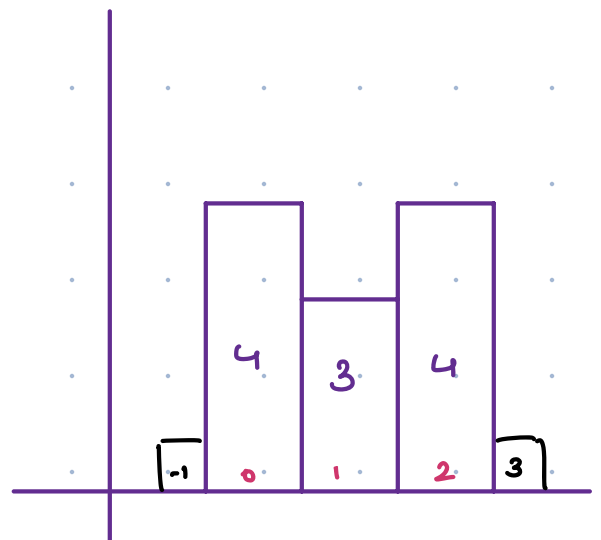
SC: O(n)



P₁ = -1 -1 1

P₂ = 1 -1 -1

width = 1 -1 -3



P₁ = -1 -1 1

P₂ = 1 3 3

width = 1 3 1

Q3 Given an integer array with distinct elements, for all the subarrays, find (max-min) & return its sum as the answer

$$A[] = \{ \overset{0}{2}, \overset{1}{5}, \overset{2}{3} \}$$

<u>subarr</u>	<u>Max-min</u>
[0 0] = {2}	2 - 2 = 0
[0 1] = {2, 5}	5 - 2 = 3
[0 2] = {2, 5, 3}	5 - 2 = 3
[1 1] = {5}	5 - 5 = 0
[1 2] = {5, 3}	5 - 3 = 2
[2 2] = {3}	3 - 3 = 0
Ans = 8	

$$A[] = \{ \overset{0}{1}, \overset{1}{2}, \overset{2}{3} \}$$

<u>subarr</u>	<u>max-min</u>
[0 0] = {1}	1 - 1 = 0
[0 1] = {1, 2}	2 - 1 = 1
[0 2] = {1, 2, 3}	3 - 1 = 2
[1 1] = {2}	2 - 2 = 0
[1 2] = {2, 3}	3 - 2 = 1
[2 2] = {3}	3 - 3 = 0
<u>Ans = 4</u>	

BF $\rightarrow$ Calculate max & min for all subarrays & find their difference
Add it to global answer

```
for (i=0; i<n; i++)
```

```
    for (j=i; j<n; j++) {
```

```
        max = -∞
```

```
        min = +∞
```

```
        for (int k=i; k≤j; k++) {
```

```
            max = Math.max(max, A[k]);
```

```
            min = Math.min(min, A[k]);
```

```
        int diff = max - min;
```

```
        ans += diff;
```

TC: $O(n^3)$
SC: $O(1)$

* Contribution Technique

$$\text{Ans} = \sum_{i=0}^{n-1} \text{contribution of } A[i]$$

$$\text{contribution of } A[i] = A[i] * \left(\begin{array}{l} \# \text{ no. of subarrays} \\ \text{where } A[i] \text{ is} \\ \text{maximum} \end{array} - \begin{array}{l} \# \text{ of subarrays} \\ \text{where } A[i] \text{ is} \\ \text{minimum} \end{array} \right)$$

$$A[] = \{2, 5, 3\}$$

subarr

$$[0 \ 0] = \{2\}$$

$$[0 \ 1] = \{2, 5\}$$

$$[0 \ 2] = \{2, 5, 3\}$$

$$[1 \ 1] = \{5\}$$

$$[1 \ 2] = \{5, 3\}$$

$$[2 \ 2] = \{3\}$$

Max-min

$$2 - 2 = 0$$

$$5 - 2 = 3$$

$$5 - 2 = 3$$

$$5 - 5 = 0$$

$$5 - 3 = 2$$

$$3 - 3 = 0$$

$$\underline{\underline{\text{Ans} = 8}}$$

contribution of $A[i]$

$$2 * (1 - 3) = -4$$

$$5 * (4 - 1) = 15$$

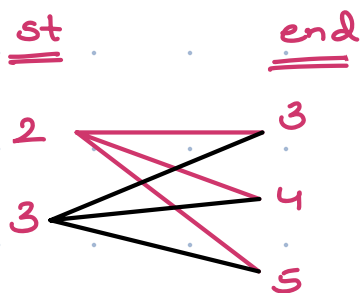
$$3 * (1 - 2) = -3$$

$$\underline{\underline{8}}$$

Q1. In how many subarrays, will 5 act as maximum?

$$A[] = \{1, 8, 3, 5, 4, 2, 11, 7\}$$

$j \longleftarrow i \longrightarrow k$



$$\underline{\underline{\text{Ans} = 6}}$$

j = next greater ele idx on LHS

k = next greater ele idx on RHS

$$\text{st}[j+1 \ i] = i - j$$

$$\text{end}[i \ k-1] = k - i$$

of subarrays

where $\frac{d^2}{dt^2}$ etc

$$= (i-j) + (k-i)$$

is maximum

01. In how many subarrays, will S act as minimum?

$A[] = \{ 1, 8, 8, 5, 9, 2, 11, 7 \}$
 p ← i → q

```
st      end
```

1 3

2 4

3

$p =$ next smaller ele on LHS

q = next smaller ele on RHS

$$\text{st}(p+1, i) = i - p$$

$$\text{end } [i \quad q-1] = q - i$$

of subarrays

where i will

act as minimum

$$= (i - p) * (q - i)$$

$$\text{Ans} = \sum_{i=0}^{n-1} A[i] * ((i - \text{greaterLeft}[i] * \text{greaterRight}[i] - 1) - (i - \text{smallerLeft}[i] * \text{smallerRight}[i] - i))$$

Final sol